

Assignment 3

August 20, 2026

1 Part 1

We're given $V(x) = B \left(1 - 4 \frac{x^2}{D^2}\right)$

We know that force is given by $F = -\frac{dV}{dx}$
taking derivative:

$$\frac{dV}{dx} = \frac{d}{dx} \left[B - \frac{4Bx^2}{D^2} \right] = -\frac{8Bx}{D^2}$$

therefore, force is:

$$F = - \left(-\frac{8Bx}{D^2} \right) = \frac{8Bx}{D^2}$$

Applying Newton's second law of motion, $F = ma \implies F = m\ddot{x}$:

$$m\ddot{x} = \frac{8Bx}{D^2} \implies \ddot{x} = \left(\frac{8B}{mD^2} \right) x$$

comparing to $\ddot{x} = \Omega^2 x$,

$$\Omega^2 = \frac{8B}{mD^2}$$

$$\Omega = \sqrt{\frac{8B}{mD^2}}$$

2 Part 2

this path represents a forced or fake motion pushed through an upside-down barrier instead of a natural motion that follows Newton's laws.

3 Part 3

kinetic energy is:

$$\frac{1}{2} m \dot{x}^2$$

The action S over a time interval from 0 to T is the integral of the Lagrangian $L = T - V$:

$$S = \int_0^T \left(\frac{1}{2} m \dot{x}^2 - V(x) \right) dt$$

Differentiating $x(t) = -\frac{D}{2} \cos\left(\frac{\pi t}{T}\right)$ with respect to time t :

$$\dot{x}(t) = \frac{D\pi}{2T} \sin\left(\frac{\pi t}{T}\right)$$

Kinetic Energy:

$$\frac{1}{2} m \dot{x}^2 = \frac{1}{2} m \left(\frac{D\pi}{2T} \sin\left(\frac{\pi t}{T}\right) \right)^2 = \frac{mD^2\pi^2}{8T^2} \sin^2\left(\frac{\pi t}{T}\right)$$

Potential Energy:

substituting $x(t) = -\frac{D}{2} \cos\left(\frac{\pi t}{T}\right)$ into $V(x)$:

$$V(t) = B \left(1 - \frac{4}{D^2} \left[-\frac{D}{2} \cos\left(\frac{\pi t}{T}\right) \right]^2 \right) = B \left(1 - \cos^2\left(\frac{\pi t}{T}\right) \right) = B \sin^2\left(\frac{\pi t}{T}\right)$$

integrating:

$$S = \int_0^T \left[\frac{mD^2\pi^2}{8T^2} \sin^2\left(\frac{\pi t}{T}\right) - B \sin^2\left(\frac{\pi t}{T}\right) \right] dt$$

$$S = \left(\frac{mD^2\pi^2}{8T^2} - B \right) \int_0^T \sin^2\left(\frac{\pi t}{T}\right) dt$$

using the formula $\int_0^T \sin^2\left(\frac{\pi t}{T}\right) dt = \frac{T}{2}$:

$$S = \left(\frac{mD^2\pi^2}{8T^2} - B \right) \frac{T}{2} = \frac{mD^2\pi^2}{16T} - \frac{BT}{2}$$

Matching $S = \frac{C_1}{T} - C_2 T$:

$$C_1 = \frac{mD^2\pi^2}{16}, \quad C_2 = \frac{B}{2}$$

4 Part 4

the most probable tunnelling time T_0 requires the action to be stationary with respect to T :

$$\left. \frac{dS}{dT} \right|_{T=T_0} = 0$$

Differentiating S with respect to T :

$$\frac{dS}{dT} = -\frac{C_1}{T^2} - C_2$$

at $T = T_0$,

$$-\frac{C_1}{T_0^2} - C_2 = 0 \implies T_0^2 = -\frac{C_1}{C_2}$$

T_0^2 is negative meaning T_0 must be imaginary:

$$T_0 = \pm i \sqrt{\frac{C_1}{C_2}}$$

where $C_3 = \sqrt{\frac{C_1}{C_2}}$

Substituting $C_1 = \frac{mD^2\pi^2}{16}$ and $C_2 = \frac{B}{2}$ into C_3 :

$$C_3 = \sqrt{\frac{\frac{mD^2\pi^2}{16}}{\frac{B}{2}}} = \sqrt{\frac{2mD^2\pi^2}{16B}} = \frac{D\pi}{4} \sqrt{\frac{m}{2B}}$$

5 Part 5

Substituting $T = -iC_3$ into $S = \frac{C_1}{T} - C_2T$ ($C_1 = C_2C_3^2$):

$$S = \frac{C_2C_3^2}{-iC_3} - C_2(-iC_3)$$

$$S = iC_2C_3 + iC_2C_3 = 2iC_2C_3$$

6 Part 6

substituting $C_2 = \frac{B}{2}$ and $C_3 = \frac{D\pi}{4} \sqrt{\frac{m}{2B}}$ into our expression for S :

$$S = 2i \left(\frac{B}{2} \right) \left(\frac{D\pi}{4} \sqrt{\frac{m}{2B}} \right)$$

$$S = i \frac{BD\pi}{4} \sqrt{\frac{m}{2B}} = i \frac{\pi D}{4} \sqrt{\frac{mB}{2}}$$

Assuming that only the path with tunnelling time T_0 contributes, the tunnelling probability P is proportional to $e^{iS/\hbar}$

$$P \propto e^{i \frac{\pi D}{4} \sqrt{\frac{mB}{2}}}$$